

Technological Institute of the Philippines
Manila Campus

PoundCare: A Digital Portal for Animal Pound Management and Pet Adoption

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I. Project Title

PoundCare: A Digital Portal for Animal Pound Management and Pet Adoption

II. Introduction

Rising stray animal populations put pressure on shelter resources and animal welfare, and this pressure is what platforms like PawFect Home try to relieve (Rallos et al., 2025). Communities also struggle because there is no comprehensive, organized system for managing strays, and user-friendly, well-integrated processes are missing (Managt Sci Tech, 2024). A survey of shelters in Portugal found that few commercial shelter management programs exist, and they cannot extract useful data such as length of stay (Marques et al., 2023). In the Philippines, LGUs impound strays for at least three days, but there is no centralized stray database and each LGU keeps its own impound logs (Respicio, 2025). This weakens lost-pet recovery: an analysis of 17 government-funded shelters showed that 43% of microchipped dogs and 78% of microchipped cats were still not returned home, and a chip is only as useful as the owner information linked to it (Hass, 2024). The root cause is that pounds rely on disconnected, non-standardized records and limited tools, so animal, vaccination, and owner information is never kept in one searchable place. Existing solutions include PawsitivelyRescued, which tracks intakes, microchip details, vaccinations, adoptions, and reminders for shelters in India, and PawFect Home, a Philippine web platform for stray reporting, sheltering, and pet adoption. Based on the reviewed studies, none of them combines pound record digitization with vaccination tracking, public adoption applications, and missing-pet alerts for Philippine city pounds.

City pounds still record animals in paper logbooks, which are easy to lose, damage, or misread. Manual records make it slow to find an animal's intake date, health status, or vaccination history. Owners of lost pets usually have to visit the pound in person, and there is no easy way to know whether their pet is there. Adoptable animals are seldom seen by the public, so many stay in the pound longer than necessary. Because LGUs keep separate impound logs, city and national monitoring of strays is difficult. A digital system would give staff accurate records and give citizens a clear way to adopt or find a lost pet.

A web-based portal replaces paper logbooks with one online database that staff can update and search anytime. Staff can log new rescues, update vaccination records, and review adoption applications in one place. Citizens can browse photos of adoptable pets and submit applications from home, without visiting the pound first. Missing-pet alerts let owners post details and compare them with animals recorded in the pound. Because it runs in a browser, it needs no special installation and works on both computers and phones.

III. Statement of the Problem

Animal pounds and shelters handle many types of information such as records of rescued animals, vaccination details, adoption requests, and lost or missing pet reports. According to Cachuela et al. (2025), manual and paper-based recordkeeping can make it harder to organize records, find information quickly, and keep data updated. A Philippine study in Alaminos City conducted by Morales et al. (2026) also found that the older methods, such as printed posters, social media posts, and paper-based checking, can make it harder to manage lost pets, impounded animals, and adoption activities.

These studies show that online systems can help make these processes easier. Sarmiento et al. (2025) developed an online platform for reporting stray animals, managing shelter records, and handling pet adoption. Another study in Cebu City found that a web and mobile application pet adoption system made the adoption process more accessible for animal shelters, pet owners, and adopters (Campanilla et al. 2022).

Because of these problems, the proposed system aims to provide one online platform where animal pound staff can manage animal records and where citizens can view pets available for adoptions, send adoption applications, and report missing pets.

Specifically, PoundCare aims to answer the following questions:

1. How can the system make it easier to record and update animal information, vaccination records, and adoption status?
2. How can the system help citizens easily view pets available for adoption and submit adoption applications online?
3. How can the system help citizens report missing pets and check if their pets may have been brought to the animal pound?

IV. Objectives

General Objective

To develop a web-based digital portal that replaces paper logbooks at local city animal pounds, improving record management, animal welfare, and public access to adoption and lost-pet services.

Specific Objectives

1. Design record management that allows shelter staff to log newly rescued animals, including their details, and intake dates, in a centralized and searchable database.
2. Implement a vaccination and health tracking feature that lets staff update and monitor each animal's immunization records accurately and on schedule.
3. Develop an online adoption system where staff can review, approve, or reject adoption applications submitted by citizens, with the status of each application tracked.

V. Target Users

PoundCare is intended for the people who manage local animal pounds and the citizens who use their services. Its main users are:

- **Pound staff:** They record newly rescued animals, including photo, breed, age, sex, condition, and update each animal's vaccination records and status. They also review missing pet reports and compare them with the animals in the pound.
- **City veterinary office personnel:** Checks the work and watches the data. They review adoption applications and monitor pound records, vaccinations, and adoptions.
- **Prospective Adopters:** They browse photos and profiles of adoptable animals, submit adoption applications, and track the status of their applications.
- **Pet owners with lost pets:** Citizens whose pets have gone missing. They post missing pet reports with photos, the last known location, and contact details and they check if their pet is in the pound.

- **System administrators:** Manages user accounts and roles and they can review system logs to see who changed what.

VI. Proposed Features

PoundCare will include the following features:

- **User Authentication** - allows staff and citizens to securely register and login to the system.
- **Animal Record Management** - staff can create, view, update, and delete records of rescued animals, including photos, breed, age, sex, condition, vaccination details, and status.
- **Adoptable Pet Listing** - citizens can browse available animals and view their complete profiles and photos
- **Adoption Application Management** - users can submit adoption applications online, while staff can review and update the application status.
- **Adoptable Pet Reporting** - citizens can submit missing pet reports containing pet details, photos, last known location, and contact information.
- **Missing Pet Monitoring** - staff and users can view reported missing pets and compare them with animals brought to the pound.
- Search and Filter - users can search and filter animals by category, breed, sex, age, or adoption status.
- **Status Management** - staff can update animal status such as Rescued, Available for Adoption, Pending Adoption, Adopted, or Reunited with Owner.
- **Dashboard** - displays summaries such as the number of rescued animals, available animals, adopted pets, missing pet reports, and pending adoption applications.
- **Reports** - allows staff to generate or view records of animal intake, adoptions, vaccination information, and missing pet cases.

VII. ReactJS Implementation

React (built with Vite) will be used to build PoundCare's front end as a single-page application. Staff and citizens will see different pages depending on their role, and all data will come from the Laravel API.

UI Components - The interface will be built from reusable components such as Navbar, AnimalCard, AnimalForm, StatusBadge, ApplicationTable, MissingPetCard, DashboardCard, and ProtectedRoute. These will be assembled into pages: Home and Adoptable Pet Listing, Pet Profile, Login/Register, Staff Dashboard, Animal Records, Adoption Applications, Missing Pet Reports, and Reports. React Router will handle navigation, and ProtectedRoute will keep staff-only pages away from citizens.

Forms - All forms like registration, login, animal record, adoption application, and missing pet report, will be controlled components, which means each input's value is kept in React state. Photo uploads will be sent with FormData. The forms will run a basic checklist for empty required fields before submitting, and any validation errors returned by Laravel will be shown beside the matching fields.

State Management - Component-level data such as form values, search text, and loading or error flags will use useState. Data that many components need, such as the logged-in user, role, and token, will be stored in an AuthContext using the Context API when a page opens or when the filters change.

Events - Typing inputs will trigger onChange handlers to update state, and forms will use onSubmit. Buttons such as Approve or Reject Application, Update Status, and Delete will use onClick, with a confirmation prompt before deleting a record. Search and filter controls will refresh the animal list whenever their values change.

API Requests - Axios will send all requests through one shared instance with the Laravel base URL, and an interceptor will attach the login token to every request. GET requests will fetch animals, applications, missing pet reports, dashboard summaries, and reports, with search and filter values sent as query parameters. POST, PUT/PATCH, and DELETE requests will create, update, and remove records. If the API returns a 401 error, the user will be sent back to the login page.

VIII. Laravel Implementation

Backend Logic

Controllers such as authController, AnimalController, AdoptionApplicationController, MissingPetReportController, DashboardController, and ReportController will hold the system's logic. Laravel Sanctum will handle token-based login, and role middleware will separate staff-only actions, such as managing animal records, updating statuses, reviewing applications, from citizen actions like applying for adoption and reporting missing pets. Business rules will include allowing applications only for animals marked Available for Adoption, and changing an animal's status when an application is approved.

API Endpoints

METHOD	ENDPOINT	PURPOSE
POST	/api/register, /api/login, /api/logout	Authentication
GET	/api/animals, /api/animals/{id}	List with search/filter and view animals
POST, PUT, DELETE	/api/animals, /api/animals/{id}	Staff manage animal records
PATCH	/api/animals/{id}/status	Staff update animal status
POST	/api/adoption-applications	Citizen submits an application
GET	/api/adoption-applications	Staff view all, citizens view their own
PATCH	/api/adoption-applications/{id}	Staff can approve or reject

POST, GET	/api/missing-pets	Submit and view missing pet reports
GET	/api/dashboard, /api/reports	Summaries and reports for staff

Validation

Form Request classes will validate incoming data before it reaches the database. Examples include required fields, a unique and valid email, an accepted list of status values, and image type and size limits for photos. If validation fails, Laravel will return a 422 response with error messages that React displays to the user.

Database Operations

Eloquent Models such as User, Animal, AdoptionApplication, and MissingPetReport, will perform the create, read, update, and delete operations on MySQL, with relationships defined between them. Migrations will create the tables, and query filters will support searching by category, breed, sex, age, and adoption status. The dashboard will use count queries to summarize animals, adoptions, missing pet reports, and pending applications. Database transactions will be used when several changes must succeed together, such as approving an application and updating the animal's status.

IX. Database - RC

PoundCare will have the following main data to be stored:

- **User Accounts and Roles**. There will be login credentials to be able to access the website, and each user account will have an assigned user role such as citizens, pound staff, veterinary personnel, and system administrators, which will have access to different pages in the website.
- **Impounded Animal Records**. This will include detailed profiles of each rescued animal, including its breed, species, sex, physical condition, and health logs such as vaccines and medications. It will also include the adoption status.
- **Medical & Vaccination Records.** Health logs of the vaccines, medicines, health checkups, and scheduled vet visits of the rescued animals.
- **System Logs and Audit Trails**. This will log any activities made by staff of the impound and veterinary personnel to track accountability for any animal status updates, application processing, and data modifications.
- **Adoption Application Records**. This will consist of forms filled out by citizens for adopting a pet. It will store users' names, age, sex, income, house pictures, and any other field required for adopting a pet.
- **Missing Pet Records**- Users will create a report for their missing pets and it will be stored here, including its name, sex, age, picture, date the pet was lost, and the location where their pet was last seen.

The major table entities are the following:

A. Users Table
   - id(PK, Auto Incement)
   - first_name(VARCHAR)
   - last_name(VARCHAR)
   - email(VARCHAR, unique)
   - password(VARCHAR, Hashed)
   - phone_number(VARCHAR)
   - address(VARCHAR)
   - role (ENUM: 'citizen', 'pound_staff', 'veterinary_staff', 'admin')
   - created_at/updated_at(TIMESTAMP)

B. Animals Table
   - id (Primary Key, Auto Increment)
   - animal_code (VARCHAR, Unique) — e.g., PC-2026-001
   - name (VARCHAR)
   - species (ENUM: 'Dog', 'Cat', 'Other')
   - breed (VARCHAR)
   - age (VARCHAR)
   - sex (ENUM: 'Male', 'Female')
   - photo_url (VARCHAR)
   - status (ENUM: 'Rescued', 'Available for Adoption', 'Pending Adoption', 'Adopted', 'Reunited with Owner')
   - created_at / updated_at (TIMESTAMP)

C. Health_Logs Table
   - id (PK, BigInteger, Auto Increment)
   - animal_id (FK -> animals.id, ON DELETE CASCADE)
   - spay_status(ENUM: ‘Neutered’, ‘Not Neutered’)
   - vaccine_name (VARCHAR 150, NOT NULL) — e.g., "Anti-Rabies", "DHPP"
   - administered_date (DATE, NOT NULL)
   - next_due_date (DATE, NULLABLE)
   - administered_by (FK -> users.id, NULLABLE) — Refers to Veterinary Staff
   - remarks (TEXT, NULLABLE)
   - created_at / updated_at (TIMESTAMP)

D. Adoption_Applications
   - id (Primary Key, Auto Increment)
   - applicant_id (Foreign Key -> users.id)
   - animal_id (Foreign Key -> animals.id)
   - applicant_age (INTEGER)
   - applicant_sex (VARCHAR)
   - monthly_income (DECIMAL / VARCHAR)
   - house_pictures_url (VARCHAR)
   - reason_for_adoption (TEXT)
   - status (ENUM: 'Pending', 'Under Review', 'Approved', 'Rejected')

- reviewed_by (Foreign Key -> users.id, Nullable)
- created_at / updated_at (TIMESTAMP)

E. Missing_Pet Table
- id (Primary Key, Auto Increment)
- reporter_id (Foreign Key -> users.id)
- pet_name (VARCHAR)
- species (ENUM: 'Dog', 'Cat', 'Other')
- sex (ENUM: 'Male', 'Female')
- age (VARCHAR)
- photo_url (VARCHAR)
- date_lost (DATE)
- last_seen_location (VARCHAR)
- contact_number (VARCHAR)
- status (ENUM: 'Active', 'Reunited', 'Closed')
- created_at / updated_at (TIMESTAMP)

F. Audit_Logs Table
- id (Primary Key, Auto Increment)
- user_id (Foreign Key -> users.id)
- action (VARCHAR) — e.g., "Updated Animal Status"
- auditable_id (BigInteger) — FK pointing to the specific ID of the modified record.
- created_at (TIMESTAMP)

X. Cloud Deployment Plan

The chosen deployment plan for this website will be made purely without any infrastructure costs as the development group will utilize a modern serverless cloud deployment model. The website will be using Vercel as the primary hosting platform due to its fast deployments, zero-cost tier, and native support for front-end frameworks. Furthermore, Vercel's frontend and API hosting will ensure high availability, speed, and reliability of the system.

The user interface rendering and application logic will be deployed directly on Vercel's Content Delivery Network, as it connects directly to the project's official GitHub repository, which will enable an automated CI/CD pipeline wherein every code change will trigger an instant build and deployment.

There will also be use of Aiven for MySQL, which will be connected to Vercel. Aiven was chosen as it provides a fully managed and free MySQL database tier for 1 GB of dedicated storage, automated backups, and encrypted database connectivity. As Vercel runs serverless, the connection environment variable will be securely configured inside the Vercel Project Settings vault, which allows backend routes or API calls to encrypt TLS/SSL connections to the MySQL database.

XI. Expected Output

PoundCare, a web application that would be replacing the logbooks that city animal pounds currently use on paper with a single, online system. The front end is built using React (with Vite) and the back end is built using Laravel. They have a REST API to communicate between each other and a MySQL database to store the data. It is implemented on the web and runs in a web browser, so users have no need to install anything, and can access it on a computer or a phone.

The staff side is a secure site for the staff of the pound, the City's veterinary office staff, and system administrators. Pound staff can photograph and document new animals found, breed, age, sex and condition, and maintain up-to-date information on each animal's vaccination status and history. Veteran staff can access applications for adoption and keep track of pound information. Administrators can add and remove user roles and accounts and check the audit logs for changes made by users. A dashboard displays the number of animals rescued, available animals, animals adopted, missing pet reports and pending applications; staff can access reports on intakes, adoptions, vaccinations and missing pet cases.

The public side is for people who register as a citizen. Potential adopters have the ability to view photos and complete profiles of available animals, to search and filter animals, to submit adoption applications online and to monitor the status of their application. To help lost pets find their way back to their owners, pet owners can post missing pet reports with photos, the last location that the pet was seen and contact information, and also determine if their pet is in the pound.

With the completion of the project, this will consist of the working front end for React, the API for Laravel with Sanctum login, and the MySQL database and tables for users, animals, health logs, adoption applications, missing pet reports and audit logs. It will also feature the live app deployed online using Vercel and the database will be served by Aiven. These create searchable and accurate records for pound staff and a pet-adoption program for citizens.

XII. References

Cachuela, S. J., Matias, D. M., Jose, R. J., & Altura, J. (2025). Navigating veterinary practice in the digital age: Implementing a web-based information management system at Animals' Choice Clinic. Psychology and Education: A Multidisciplinary Journal. https://doi.org/10.70838/pemj.370506

Campanilla, B. S., Etcuban, J. O., Maghanoy, A. P., Nacua, P. A. P., & Galamiton, N. S. (2022). Pet adoption app to free animal shelters. Journal of Positive School Psychology. Mail Journal PPW. https://journalppw.com/index.php/jpsp/article/view/10869

Hass. (2024, May 24). *Microchipped pets are THREE TIMES more likely to get home*. HASS. Retrieved September 30, 2026, from https://www.humananimalsupportservices.org/uncategorized/new-analysis-pets-with-microchips-are-three-times-more-likely-to-get-home/

Managt Sci Tech, I. R. journa. (2024). Technology-Driven Animal Shelter Management: The PawsitivelyRescued DBMS Sunidhi Sudhir E. K. Isarasolutions. https://doi.org/10.32804/IRJMST

Marques, S., Gomes-Neves, E., Baptista, C. S., Pereira, F. R., Alves-Pereira, A., Osório, P., & Müller, A. (2023). A survey on vaccination and disease occurrence in Municipal and Non-Profit animal shelters in Portugal. *Animals*, *13*(17), 2723. https://doi.org/10.3390/ani13172723

Morales, E. C., Eusebio, J. P., Ringor, V. R., Caalim, J. B., & Melendez, M. L. D. (2026). Development and evaluation of a digital pet recovery and adoption management system for Alaminos City. International Journal of Research and Innovation in Social Science. https://doi.org/10.47772/IJRISS.2026.100600300

Rallos, C. J., Molina, F., Sarmiento, H. H., Sarsua, J., Barnatia, J., Ordonio, J. M., & Pujeda, Z. D. M. (2025). *PawFect Home: an online stray animal reporting, sheltering, and pet*

adoption platform. Retrieved September 30, 2026, from https://www.ejournals.ph/article.php?id=28426

Respicio, H. (2025, June 24). *Stray dog legal definition under Philippine Animal Welfare Act — Respicio & Co.* Respicio & Co. Retrieved September 30, 2026, from https://www.lawyer-philippines.com/articles/stray-dog-legal-definition-under-philippine-animal-welfare-act

Sarmiento, H. H., Ordonio, J. M., Pujeda, Z. D. M., Molina, F., Sarsua, J., Barnatia, J., & Rallos, C. J. (2025). PawFect Home: An online stray animal reporting, sheltering, and pet adoption platform. Psychology and Education: A Multidisciplinary Journal. https://doi.org/10.70838/pemj.400410